

Azure App Services Deployment Guide

Complete setup, environment configuration, and runtime risk register for AtlasRide on Azure

Prepared by AtlasRide Engineering · 28 April 2026

AtlasRide is a full-stack Node.js / React application. The server is a compiled CommonJS bundle (`dist/index.cjs`) that serves both the REST API and the pre-built React frontend. In production it connects to a PostgreSQL database and communicates with Stripe, Twilio, Azure Maps, and Microsoft Entra External ID.

- **Runtime:** Node.js 20 (LTS)
- **Start command:** `npm run start` | `node dist/index.cjs`
- **Port:** Reads `process.env.PORT` (Azure sets this automatically)
- **Database:** PostgreSQL — Azure Database for PostgreSQL Flexible Server recommended
- **Session store:** PostgreSQL-backed via `connect-pg-simple`
- **File uploads:** Local disk (see Risk #4 below for Azure caveat)

- Azure App Service Plan (Linux, Node 20 LTS)
- Azure App Service (Web App)
- Azure Database for PostgreSQL Flexible Server
- Optional: Azure Blob Storage (for persistent file uploads)

Step 2 — Build the Application

Run the following from the project root before deploying:

This produces `dist/index.cjs` and `dist/public/`. Both must be deployed to Azure.

Step 3 — Deploy Code to Azure

- Via GitHub Actions: connect your repo in Azure Deployment Center
- Via ZIP Deploy: zip the project (including `dist/`) and upload via Kudu
- Via Azure CLI: `az webapp deploy --src-path ./dist`

Set all of these in Azure Portal !' App Service !' Configuration !' Application Settings. Without these the application will crash on startup.

Variable Name	Value / Description	Required
NODE_ENV	production	YES
DATABASE_URL	postgresql://user:password@host/dbname?sslmode=require	YES
SESSION_SECRET	Any 32+ character random string (use a password generator)	YES
AZURE_MAPS_KEY	Your Azure Maps subscription key	YES
TWILIO_ACCOUNT_SID	Your Twilio Account SID (ACxxxxxxxxxxxxxxxx)	YES
TWILIO_AUTH_TOKEN	Your Twilio Auth Token	YES
TWILIO_MESSAGING_SERVICE_SID	Your Twilio Messaging Service SID	YES
ENTRA_CLIENT_SECRET	Microsoft Entra External ID client secret (for email OTP)	YES
MAPBOX_ACCESS_TOKEN	Your Mapbox public access token	Optional
WEBSITE_HOSTNAME	Auto-set by Azure — do not override	Optional

IMPORTANT: DATABASE_URL must end with ?sslmode=require when connecting to Azure Database for PostgreSQL. Without this the connection will be refused by Azure.

4. Database Setup on Azure

```
DATABASE_URL="postgresql://user:pass@host/db?sslmode=require" npm run db:push
```

Set DATABASE_URL to your Azure PostgreSQL connection string, then:

After provisioning your Azure PostgreSQL server and creating a database, you must run the schema migration from your local environment (or CI pipeline) pointing at the Azure database. The db:push command above will create it automatically for you. The app connects to a session table in user session storage. The db:push command above will create it automatically for you. The app connects to a session table in user session storage.

postgresql://USERNAME:PASSWORD@SERVER_NAME.postgres.database.azure.com/DATABASE_NAME?sslmode=require

4c. Connection String Format

The following issues were identified by a code audit. All critical items must be resolved before go-live.

Severity	Issue	Detail	Resolution
CRITICAL	Missing <code>SESSION_SECRET</code>	The app uses <code>session.secret</code> with a non-null assertion (!). If <code>SESSION_SECRET</code> is not set in Azure App Settings the server crashes on startup.	Set <code>SESSION_SECRET</code> in Azure App Settings. Use a 40+ character random string.
CRITICAL	Missing <code>DATABASE_URL</code>	Without <code>DATABASE_URL</code> the PostgreSQL pool fails to initialise. The server will start but every DB query crashes with an unhelpful error.	Set <code>DATABASE_URL</code> pointing to your Azure PostgreSQL server with <code>?sslmode=require</code> .
CRITICAL	Sessions table absent	<code>connect-pg-simple</code> is configured with <code>createTableIfMissing: false</code> . If the sessions table does not exist in the database the app will throw an error on login attempt.	Run <code>npm run db:push</code> against the Azure DB before first deployment.
MEDIUM	Replit OIDC on Azure	The <code>/api/login</code> route uses Replit OAuth (openid-client). <code>REPL_ID</code> and <code>ISSUER_URL</code> are Replit-specific. This route will fail silently on Azure.	Users should log in via email/password (<code>/api/auth/login</code>). Replit OAuth is not required and can be ignored on Azure.
MEDIUM	File uploads — ephemeral disk	Driver licence photos and profile images are written to the local filesystem (<code>uploads/</code>). Azure App Service restarts wipe the local filesystem on every deployment.	Long-term: migrate file storage to Azure Blob Storage. Short-term: uploads persist across warm restarts but not deployments.
MEDIUM	Stripe webhook URL	The webhook URL is auto-built from <code>WEBSITE_HOSTNAME</code> (set by Azure). This is correct but must be verified in your Stripe dashboard after first deployment.	Check Stripe Dashboard ! Webhooks after first deployment to confirm the URL matches your Azure domain.
LOW	<code>reusePort</code> flag	Server was configured with <code>reusePort: true</code> which is Linux-only and unsupported on some Azure configurations.	FIXED — <code>reusePort</code> is now disabled automatically when running on Azure (detected via <code>WEBSITE_HOSTNAME</code>).
LOW	Vite HMR WebSocket	Browser console shows WebSocket HMR errors in development. Not present in production.	No action needed — this only appears in the Replit development preview, not in production.

`npm run start`

In Azure Portal ! App Service ! Configuration ! General Settings, set:

Node.js Version

Always On

Enable "Always On" in App Service Configuration ! General Settings to prevent the app sleeping and losing WebSocket connections.

WebSocket Support

Enable WebSockets in App Service Configuration ! General Settings. AtlasRide uses WebSockets for real-time driver tracking and in-ride chat.

- Visit <https://YOUR-APP.azurewebsites.net/> — should show the AtlasRide login screen
- Visit <https://YOUR-APP.azurewebsites.net/api/auth/user> — should return 401 (confirms API is live)
- Create an account and log in — confirms database connection and session store are working
- Upload a profile photo — confirms file upload route is working
- Check Stripe Dashboard ! Webhooks — confirm webhook URL matches your domain

Logs

`az webapp log tail --name YOUR-APP --resource-group YOUR-RG`

View real-time logs in Azure Portal ! App Service ! Log Stream, or use Azure CLI:

Print and keep this card during setup. Tick each item as it is configured.

- | | |
|---|--|
| ☐ NODE_ENV = production | ☐ DATABASE_URL = postgresql://...?sslmode=require |
| ☐ SESSION_SECRET = (40+ random chars) | ☐ AZURE_MAPS_KEY = (Azure Maps key) |
| ☐ TWILIO_ACCOUNT_SID = ACxxx... | ☐ TWILIO_AUTH_TOKEN = (auth token) |
| ☐ TWILIO_MESSAGING_SERVICE_SID = MGxxx... | ☐ ENTRA_CLIENT_SECRET = (client secret) |
| ☐ MAPBOX_ACCESS_TOKEN = pk.xxx... (optional) | ☐ db:push run against Azure PostgreSQL |
| ☐ Startup command set to: npm run start | ☐ Node version set to 20 LTS |
| ☐ Always On: ENABLED | ☐ WebSockets: ENABLED |
| ☐ Stripe webhook URL verified in dashboard | |

Questions or issues? All deployment decisions described in this document are based on an audit of the AtlasRide codebase as of the date shown on the cover. Re-run the audit after major code changes.

